

Johan Eklund, Josh Hagedorn, Sándor Darányi

Teaching Tale Types to a Computer: A First Experiment with the Annotated Folktales Collection

<https://doi.org/10.1515/fabula-2023-0005>

Abstract: Computational motif detection in folk narratives is an unresolved problem, partly because motifs are formally fluid, and because test collections to teach machine learning algorithms are not generally available or big enough to yield robust predictions for expert confirmation. As a result, standard tale typology based on texts as motif strings renders its computational reproduction an automatic classification exercise. In this brief communication, to report work in progress we use the Support Vector Machine algorithm on the ten best populated classes of the Annotated Folktales test collection, to predict text membership in their internationally accepted categories. The classification result was evaluated using recall, precision, and F_1 scores. The F_1 score was in the range 0.8–1.0 for all the selected tale types except for type 275 (*The Race between Two Animals*), which, although its recall rate was 1.0, suffered from a low precision.

Zusammenfassung: Die computergestützte Motiverkennung in Märchenerzählungen ist ein ungelöstes Problem, sowohl weil Motive formal fließend sind als auch weil Testsammlungen zur algorithmischen Erkennung für maschinelles Lernen nicht ausreichend verfügbar oder groß genug sind, um robuste Vorhersagen für die Bestätigung durch Experten zu liefern. Daher ist die rechnerische Reproduktion der bislang gängigen Märchentypologie, die auf Texten als Motivketten basiert, unvermeidlich eine automatisierte Klassifizierungsübung. In diesem Bericht über unsere laufende Arbeit wenden wir deshalb einen Support Vector Machine-Algorithmus auf die zehn am besten annotierten Klassen der Annotated Folktales-Testsammlung an, um die Zugehörigkeit von Texten zu ihren international anerkannten Kategorien vorherzusagen. Das Klassifizierungsergebnis wurde anhand von Recall-, Präzisions- und F1-Scores bewertet. Die F1-Werte lagen zwi-

Sándor Darányi, Professor Emeritus at Högskolan i Borås, Faculty of Librarianship, Information, Education and IT, Swedish School of Library and Information Science, Borås, Sweden. E-mail: sandor.daranyi@hb.se

Johan Eklund, Senior Lecturer, Högskolan i Borås, Faculty of Librarianship, Information, Education and IT, Swedish School of Library and Information Science, Borås Sweden. E-Mail: johan.eklund@hb.se

Josh Hagedorn, Independent researcher, Högskolan i Borås, Faculty of Librarianship, Information, Education and IT, Grand Rapids, USA. E-mail: josh.hagedorn@gmail.com

schen 0.8 und 1.0 für alle ausgewählten Märchentypen, mit Ausnahme der Klasse 275 (*The Race between Two Animals*), die eine Recall-Rate von 1.0, aber eine geringe Präzision aufwies.

1 Introduction

Ever since the concept of a motif was introduced some two hundred years ago, the quest to identify elements of content above the word level has been a standard preoccupation in literary science (Frenzel 1992; Seigneuret 1988). There, a motif stands for a recurrent theme, whereas in musicology a motif is considered “the smallest structural unit possessing thematic identity” (White 1976, 26–27). In the field of folktale research, Stith Thompson defined motifs as “the smallest element in a tale having a power to persist in tradition” (Thompson 1977, 415).

The overlap between these definitions suggests that robust higher-order content units as narrative building blocks exist, yet their automatic extraction by computational means has eluded folk narrative studies so far. One reason for this shortcoming is that in *Thompson’s Motif Index of Folk Literature* (Thompson 1951) alone, over 45,000 motifs are listed on a global scale, but other regional motif indexes exist which would inflate that number, if their motifs were harmonized with the TMI. To use machine learning for motif identification, we need common reference datasets which enable research teams to replicate one another’s results. Storytellers from across times and cultures have exercised creativity in their reuse of semantic content, requiring taxonomists and computational folklorists to account for a considerable degree of fluidity in their identification of recurrent content. This less than clear-cut nature of motifs is an obstacle when it comes to their detection by pattern recognition, yet it is crucial for problems of classification. As Propp noted, “[i]f we are incapable of breaking the folktale down to its components, then we shall be unable to make a correct comparison” (Propp 1958, 14).

A consequent difficulty is manifest in the motif-based typology of tales, such as the Aarne-Thompson-Uther classification of international folktales (ATU) (Uther 2004). As these tale types go back to strings of flexibly defined motifs, to assess the belonging of texts to such types equals their classification according to sequences of specific semantic content, where, however, such content can be expressed on word phrase, sentence, or paragraph level, perhaps even above. Therefore, here we report on initial attempts to “learn” from the expert classifications of the ATU using support vector machines (SVM) in a way that allows for reproducible research and collaboration with others in the field.

The structure of this article is as follows. In section 2, we briefly discuss related research, including the availability of databases and datasets, construction of the Annotated Folktales (AFT) test collection, then going over to computational methods used in folktale research, especially examples relevant to our current attempt. In section 3 we discuss preprocessing and experiment design, leading to our results in section 4, and their discussion with an eye on future directions in section 5. Tale type composition of the AFT is displayed in the Appendix.

2 Related research

As computational folktale research needs open access datasets for reproducibility of the results, we briefly look at their availability next.

2.1 Databases and datasets

Progress in computational folkloristics requires that results be replicable. To this end we sought open access datasets of ATU-annotated tales in English, but could not identify suitable candidates on GitHub¹, Kaggle² or Google³, although websites with separate tale collections are available. Neither could we find the big folklore data anticipated by Tangherlini & Leonard (2013) and Tangherlini (2016). Based on Meder (2010) and Ilyefalvi (2018), the largest databases seem to be the Dutch Folktale Database of the Meertens Institute, and the Danish Folklore Archive's Tang Kristensen Collection, the former in the magnitude of around 50,000 texts, the latter at around 34,000 texts (Tangherlini/Leonard 2013). Other important databases exist (Berezkin 2017), but are either beyond public access, or in their original languages only, or both. The notable exception is the Meertens Institute whose texts are in Dutch and Frisian plus several local dialects but can be read in English translation as well.

Other researchers who have shared their data as supporting material for their articles include for instance Bortolini et al. (2017), Silva & Tehrani (2016), Tehrani (2013), and Tehrani, Nguyen, & Roos (2016). Also, (Declerck/Schäfer 2017) and (Declerck/Kostova/Schäfer 2017) report that recently, a large amount of ATU data have been made available online by the Multilingual Folk Tale Database (MFTD)⁴,

¹ <https://github.com/awesomedata/awesome-public-datasets> (September 4, 2022).

² <https://www.kaggle.com/datasets> (September 4, 2022).

³ <https://datasetsearch.research.google.com/> (September 4, 2022).

⁴ <https://www.mftd.org> (September, 4 2022).

also offering annotation facilities for tales in multilingual versions. We found only a single recent study (Lô/Boer/Aart 2020) which published a corresponding tale corpus to promote reproducibility, albeit without ATU type labels.⁵

2.2 From the Ashliman's Folktexts collection to the Annotated Folktales dataset

From the ATU-annotated tale collections publicly available on the internet, the most promising candidate was Ashliman's Folktexts collection. The Folktexts site⁶ has been populated and maintained since 1996 by D. L. Ashliman, who kindly agreed to donate his collection to the interested research communities.⁷ While other sites may sport a more lavish design, this one is the largest and most extensively annotated. It serves as a respected scholarly resource for folklorists, with a large and curated set of tale texts. Whereas our dataset contains only tales from pages with clear ATU annotations ($n = 214$ pages), the total content of the website is much larger ($n = 370$ pages), containing various creation myths, stories of changelings, Faust legends, and Christiansen's tale types (Christiansen 1992). However, it is the ATU annotation that makes this corpus particularly valuable as a potential training dataset for classification methods.

Despite the richness of this resource, it has not frequently been used in folklore research as a larger corpus. Some previous studies reference it, yet these often only include a smaller portion of the entire set of texts (Reiter/Frank/Hellwig 2014). To the best of our knowledge, none of the published studies provided open access to the data.

Having converted the Ashliman's Folktexts collection into a new format enabling data science applications, elsewhere we gave detailed specifics of the Annotated Folktales open access dataset (Hagedorn/Darányi 2022). The AFT consists of 1,518 texts annotated with 182 ATU tale types. Distribution of the tale types according to ATU genres is shown in Appendix.

2.3 Computational methods used in folktale research

As our effort was not concerned with the structural analysis of folk narratives, this brief overview omits significant research results, such as those concerning the

⁵ <https://github.com/GossaLo/afr-neural-folktales/> (September 4, 2022).

⁶ <http://www.pitt.edu/~dash/folktexts.html> (September 4, 2022).

⁷ Personal communication per email (February 11, 2021).

automatic detection of Proppian functions (Finlayson 2016), or their use in ontology building (Declerck et al. 2017). Important extensions to these, and to our current work, exist by Declerck and colleagues (Declerck/Schäfer 2017; Declerck/Kostova/Schäfer 2017). The examples below exploit the TMI and the ATU, respectively.

In computational folkloristics (Abello/Broadwell/Tangherlini 2012), the focus is often on the evolutionary aspect of motif and/or tale type distributions, either with regard to certain tale types (Bortolini et al. 2017; Karsdorp 2016; Karsdorp/van den Bosch 2013; Silva/Tehrani 2016; Tehrani 2013), or to the geographical distribution of globally occurring narrative motifs (Thuillard/d'Huy/Berezkin/Le Quellec 2018), even inferring the presence of lost narratives (Kestemont et al. 2022). It is standard practice to use probabilistic and/or multivariate statistical methods for the analysis of binary versus non-binary matrices of events over cases, where events can be index terms, motifs, motif sequences, and so on, and cases as an umbrella term stand for documents in general, such as abstracts describing narratives (Berezkin 2015), or tale types (Uther 2004), ultimately constituting text corpora or databases. Regardless of the corpora, their regionality or the analytical units whose distributions characterize the body of texts in question, they express similarity between items in terms of distance, with more similar items forming dense groups as the outcome of mass comparison.

Previous studies have used such collections to conduct experiments using sub-corpus topic modeling (STM) by Latent Dirichlet Allocation (LDA) as a means of supervised passage exploration in partly unknown corpora (Tangherlini/Leonard 2013), cluster analysis (Thuillard/d'Huy/Berezkin/Le Quellec 2018), Principal Component Analysis (PCA) (Berezkin 2015), Labeled Latent Dirichlet Allocation (L-LDA) (Karsdorp/van den Bosch 2013), or deep learning by Recurrent Neural Networks (RNN) (Lô/Boer/Aart 2020). A promising recent direction seems to be the combination of word embeddings – increasingly condensed and geometrically located types of word meaning – with deep learning: Pompeu et al. report successful application of a Hierarchical Attention Network (HAN) for the prediction of ATU categories on a multilingual database of folk texts (Pompeu/Martins/Matos 2019).

2.4 Nonlinear document classification by SVM

Statistical learning theory is one of the fields of inferential statistics whose basis has been established by V. N. Vapnik in the late 1960s (Vapnik 1999). The goal of this theory was to specify the conditions under which it is possible to “learn” from empirical data obtained by random sampling. Learning amounts to solving a problem of model selection, e.g., finding an automatic classification matching the manual one as closely as possible.

It is a common problem in classification that a linear model will not succeed in producing a good separation between the examples. To help this, the trick is to map the data, using a nonlinear function, into a new feature space, and then use a linear classifier to arrive at a better result than before. This is done by support vector machines, which constitute the state-of-the-art for numerous problems in pattern recognition (Figure 1). SVMs are just one application of a broader, fascinating field called “kernel methods”. Kernel machines provide a modular framework that can be adapted to different tasks and domains by the choice of the kernel function and the base algorithm.

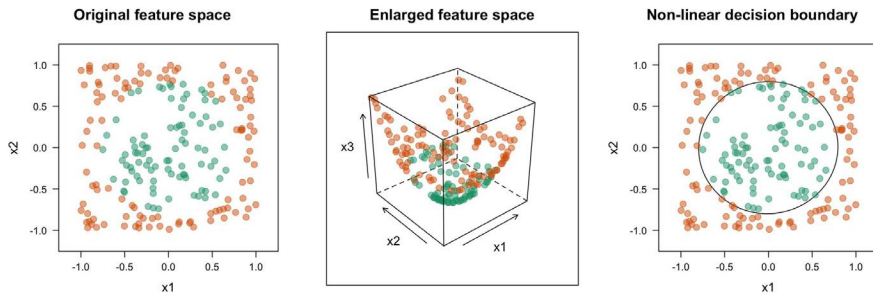


Fig. 1: Simulated nested circle data

Left: The two classes in the original (2-D) feature space

Middle: The two classes in the enlarged (3-D) feature space

Right: The decision boundary from the hard margin classifier in the enlarged feature space projected back into the original feature space. A hard margin classifier is the boundary that provides the maximum separation between the two classes (from Fig. 14.6, in Boehmke and Greenwell 2019)

Since SVM can be used for the unsupervised learning of narrative event chains (Chambers/Jurafsky 2008), it is present in the text analytical toolkit of several computational folktale research teams (Eisenberg/Yarlott/Finlayson 2016; Pajupuu/Altrov/Pajupuu 2016; Lô/Boer/Aart 2020). However, its use to detect ATU tale types is sporadic to the best of our knowledge. Inspired by (Nguyen/Trieschnigg/Meder/Theune 2012), who used an SVM with a linear kernel and L2 regularization to automatically recognize ATU-labeled folk narrative genres such as urban legends, fairy tales, jokes, and riddles in the Dutch Folktale Database with promising results, (Pompeu/Martins/Matos 2019) used the same SVM parameterization as one of two baselines to compare deep learning performance on the ATU-based classification of folktales. Nguyen, Trieschnigg & Theune (2013) was focusing on story type identification by a ranking approach and mentions in passing among others the use of SVM as well (2013, 205). Also including SVM in their methodology, Lô, Boer, & Aart (2020)

describe a three-stage experiment to find out how machine learning and natural language processing can be used to identify, analyze, and generate West African folk tales, however, outside of the ATU genre and type taxonomy. Pajupuu, Altrov, & Pajupuu (2016) use SVM for sentiment mining on Estonian folklore texts without ATU type membership.

3 Material and method

3.1 Chosen subset of the test collection

For experiment design, we used the ten most frequent tale types in the AFT. This subset had 238 texts (Table 1), which amounted to 15.68 % of the collection (Table 2). In ranking order, the ATU genres best represented in this subset were “Tales of magic (300–749)” with 111 texts (46.6 % of the subset; 7.3 % of the AFT); “Animal tales (1–299)” with 51 texts (21.4 % of the subset; 3.4 % of the AFT); “Religious tales (750–849)” with 30 texts (12.6 % of the subset; 2 % of the AFT); “Anecdotes and jokes (1200–1999)” with 26 texts (10.9 % of the subset; 1.7 % of the AFT); and “Realistic tales (Novelle) (850–999)” with 20 texts (8.4 % of the subset; 1.3 % of the AFT). The genre “Tales of the stupid ogre (giant, devil) (1000–1199)” was not represented among the ten most frequent tale types.

Tab. 1: The ten most frequent tale types in the AFT

ATU ID	Tale type name	Number of tales
275	<i>The Race between Two Animals</i> (previously <i>The Race of the Fox and the Crayfish</i>)	31
777	<i>The Wandering Jew</i>	30
1645	<i>The Treasure at Home</i>	26
510B	<i>Peau d'Asne</i> (previously <i>The Dress of Gold, of Silver, and of Stars [Cap o' Rushes]</i>)	26
500	<i>The Name of the Supernatural Helper</i>	23
510A	<i>Cinderella</i>	21
700	<i>Thumbling</i> (previously <i>Tom Thumb</i>)	21
155	<i>The Ungrateful Snake Returned to Captivity</i>	20
545B	<i>Puss in Boots</i>	20
980	<i>The Ungrateful Son</i> (previously <i>Ungrateful Son Reproved by Naive Actions of Own Son</i>)	20

Tab. 2: Genre membership of the ten most frequent tale types in the AFT

GENRES OF THE 10 MOST FRE- QUENT TYPES	275	777	1645	510B	500	510A	700	155	545B	980	NUMBER OF TEXTS	% OF SUBSET	% OF AFT
1. ANIMAL TALES (1–299)	31							20			51	21.4	3.4
2. TALES OF MAGIC (300–749)				26	23	21	21		20		111	46.6	7.3
3. RELIGIOUS TALES (750–849)		30									30	12.6	2.0
4. REALIS- TIC TALES (NOVELLE) (850–999)									20	20		8.4	1.3
5. TALES OF THE STUPID OGRE (GIANT, DEVIL) (1000–1199)													
6. ANECDOTES AND JOKES (1200–1999)			26								26	10.9	1.7
7. FORMULA TALES (2000–2399)													
AFT TOTAL											238	100	15.68

3.2 Experiment design

The subcollection of tales was split so that 80 % of the documents (190 documents) were used for training and 20 % percent of the documents (48 documents) were used for evaluation. The texts were vectorized using the TF-IDF weighting scheme and sublinear TF scaling, using words having a document frequency ≥ 2 as features. No stop word removal was performed. The classification algorithm applied was the support vector machine (C-SVM), using a linear kernel and the cost parameter set to C=1.0.

4 Results

The resulting confusion matrix is presented in Table 3. In general, the ratio of true positives is high, which shows high percentage values along the diagonal from the upper-left cell to the lower-right cell of the matrix. Regarding tale type 275 (*The Race between Two Animals*), it can be observed that texts of several other tale types (more specifically 510B, 545B, and 777) have been assigned to tale type 275, resulting in a lower precision for that tale type (see Table 4).

Tab. 3: Confusion matrix for the 10 selected tale types

Reference	155	1645	275	500	510A	510B	545B	700	777	980
155	80	0	0	0	0	0	20	0	0	0
1645	0	100	0	0	0	0	0	0	0	0
275	0	0	100	0	0	0	0	0	0	0
500	0	0	0	75	25	0	0	0	0	0
510A	0	0	0	0	100	0	0	0	0	0
510B	0	0	14	0	0	71	0	0	0	14
545B	0	0	20	0	0	0	80	0	0	0
700	0	0	0	0	0	0	0	100	0	0
777	0	0	12	0	0	0	0	0	88	0
980	0	0	0	0	0	0	0	0	0	100

Note. The numbers in the table are reported as percentage values of the row total.

The classification result in the test collection was also evaluated using recall, precision, and F_1 scores, and the performance scores are shown in Table 4. The F_1 score is in the range 0.8–1.0 for all the selected tale types except type 275 (*The Race between Two Animals*), which suffers from a low precision (although the recall is 1.0).

Tab. 4: Recall, precision, and F_1 scores for the ten selected tale types

ATU ID		Recall	Precision	F_1
155	<i>The Ungrateful Snake Returned to Captivity</i>	0.8000	1.0000	0.8889
1645	<i>The Treasure at Home</i>	1.0000	1.0000	1.0000
275	<i>The Race between Two Animals</i>	1.0000	0.4000	0.5714
500	<i>The Name of the Supernatural Helper</i>	0.7500	1.0000	0.8571
510A	<i>Cinderella</i>	1.0000	0.8000	0.8889
510B	<i>Peau d'Asne</i>	0.7143	1.0000	0.8333
545B	<i>Puss in Boots</i>	0.8000	0.8000	0.8000
700	<i>Thumbling</i>	1.0000	1.0000	1.0000
777	<i>The Wandering Jew</i>	0.8750	1.0000	0.9333
980	<i>The Ungrateful Son</i>	1.0000	0.8571	0.9231

5 Discussion and future research directions

The assessment of performance scores in the above experiment was to some extent subjective of course, but F1 scores in the range 0.8–1.0 (with one exception) is in our experience a really good result. Therefore, we plan to continue this work with cross-validation of the current results and expect future extensions of our effort to the complete AFT dataset to tell how these evaluation measures degrade with sample size. As reliability of tale type prediction with this parameterization clearly depends on the amount of teaching material, we strive toward finding out if lessons learnt, based on underlying motif chains, can be transferred to other collections. An updated release of the repository housing the AFT is also in the pipeline, including code to reproduce this analysis.

Acknowledgements: The authors are grateful to X anonymous reviewers for helpful comments on the manuscript.

6 References

- Abello, James/Broadwell, Peter/Tangherlini, Timothy R.: Computational Folkloristics. In: *Communications of the ACM* 55,7 (2012) 60–70. <https://doi.org/10.1145/2209249.2209267>.
- Berezkin, Yuri: Spread of folklore motifs as a proxy for information exchange. Contact zones and borderlines in Eurasia. In: *Trames* 19,1 (2015) 3–14. <https://doi.org/10.3176/tr.2015.1.01>.
- Berezkin, Yuri: Peopling of the New World from Data on Distributions of Folklore Motifs. In: *Maths Meets Myths. Quantitative Approaches to Ancient Narratives*, 71–89. Eds. Ralph Kenna/Máirín MacCarron/Pádraig MacCarron. Heidelberg 2017. https://doi.org/10.1007/978-3-319-39445-9_5.
- Boehmke, B./Greenwell, B. M: *Hands-On Machine Learning with R*. New York 2019. <https://doi.org/10.1201/9780367816377>.
- Bortolini, Eugenio/Pagani, Luca/Crema, Enrico R./Sarno, Stefania/Barbieri, Chiara/Boattini, Alessio/Sazzini, Marco/Silva, Sara G. da/Martini, Gessica/Metspalu, Mait/Pettener, Davide/Luiselli, Donata/Tehrani, Jamshid J.: Inferring patterns of folktale diffusion using genomic data. In: *Proceedings of the National Academy of Sciences* 114,34 (2017) 9140–9145. <https://doi.org/10.1073/pnas.1614395114>.
- Chambers, Nathanael/Jurafsky, Dan: Unsupervised learning of narrative event chains. In: *Proceedings of ACL-08: HLT*, 789–797 (June 2008).
- Christiansen, Reidar Th.: *The Migratory Legends: A Proposed List of Types with a Systematic Catalogue of the Norwegian Variants*. Helsinki 1992.
- Declerck, Thierry/Aman, Anastasia/Banzer, Martin/Macháček, Dominik/Schäfer, Lisa/Skachkova, Natalia: Multilingual Ontologies for the Representation and Processing of Folktales. In: *Proceedings of the First Workshop on Language Technology for Digital Humanities in Central and (South-)Eastern Europe* (2017) 20–23. https://doi.org/10.26615/978-954-452-046-5_003.
- Declerck, Thierry/Kostova, Antónia/Schäfer, Lisa: Towards a linked data access to folktales classified by Thompson's motifs and Aarne-Thompson-Uther's types. In: *Proceedings of Digital Humanities*

- (2017b) 1–4. https://www.dfki.de/fileadmin/user_upload/import/9028_Dh2017_LOD_TMI-ATU_final.pdf.
- Declerck, Thierry/Schäfer, Lisa: Porting past classification schemes for narratives to a linked data framework. In: *Proceedings of DATECH2017* (2017) 123–127. <https://doi.org/10.1145/3078081.3078105>.
- Eisenberg, Joshua D./Yarlott, Victor W./Finlayson, Mark A. Comparing Extant Story Classifiers. Results & New Directions. In: *Proceedings of the 7th Workshop on Computational Models of Narrative (CMN 2016)*. eds. Ben Miller/Antonio Lieto/Remi Ronfard/Stephan G. Ware/Mark A. Finlayson. Dagstuhl 2016, 1–10. <https://drops.dagstuhl.de/opus/volltexte/2016/6707/pdf/OASICS-CMN-2016-6.pdf>.
- Finlayson, Mark A.: Inferring Propp's functions from semantically annotated text. In: *Journal of American Folklore* 129,511 (2016) 55–77. <https://doi.org/10.5406/jamerfolk.129.511.0055>.
- Frenzel, Elisabeth: *Stoffe der Weltliteratur: Ein Lexikon dichtungsgeschichtlicher Längsschnitte*. Stuttgart 1992.
- Hagedorn, Josh/Darányi, Sándor: Bearing a Bag-of-Tales: An Open Corpus of Annotated Folktales for Reproducible Research. In: *Journal of Open Humanities Data* 8,16 (2022). <http://doi.org/10.5334/johd.78>.
- Ilyefalvi, Emese: The theoretical, methodological and technical issues of digital folklore databases and computational folkloristics. In: *Acta Ethnographica Hungarica* 63,1 (2018) 209–258. <https://doi.org/10.1556/022.2018.63.1.11>.
- Karsdorp, Folgert: Retelling Stories: A Computational-Evolutionary Perspective. Radboud Universiteit 2016. <https://repository.ubn.ru.nl/bitstream/handle/2066/162268/162268.pdf>.
- Karsdorp, F./Fonteyn, L.: Cultural entrenchment of folktales is encoded in language. In: *Palgrave Communications* 5,25 (2019). <https://doi.org/10.1057/s41599-019-0234-9>.
- Karsdorp, Folgert/van den Bosch, Antal: Identifying motifs in folktales using topic models. In: *Proceedings of the 22 Annual Belgian-Dutch Conference on Machine Learning* (2013) 41–49.
- Kestemont, Mike/Karsdorp, Folgert/de Bruijn, Elisabeth/Driscoll, Matthew/Kapitan, Katarzyna A./Ó Macháin, Pádraig/Sawyer, Daniel/Sleiderink, Remco/Chao, Anne: Forgotten books. The application of unseen species models to the survival of culture. In: *Science* 375,6582 (2022) 765–769. <https://doi.org/10.1126/science.abl7655>.
- Lô, Gossa/Boer, Victor de/Aart, Chris J. van: Exploring West African Folk Narrative Texts Using Machine Learning. In: *Information* 11,5 (2020) 236. <https://doi.org/10.3390/info11050236>.
- Meder, Theo: From a Dutch Folktale Database towards an International Folktale Database. In: *Fabula* 51,1–2 (2010) 6–22. <https://doi.org/10.1515/FABL.2010.003>.
- Meder, Theo/Karsdorp, Folgert/Nguyen, Dong/Theune, Mariët/Trieschnigg, Dolf/Muiser, Iwe (2016). Automatic Enrichment and Classification of Folktales in the Dutch Folktale Database. In: *The Journal of American Folklore* 129,511 (2016) 78–96. <https://doi.org/10.5406/jamerfolk.129.511.0078>.
- Nguyen, Dong/Trieschnigg, Dolf/Meder, Theo/Theune, Mariët: Automatic classification of folk narrative genres. In: *Proceedings of KONVENS 2012*. ed. Jeremy Jancsary (2012) 378–382.
- Nguyen, Dong/Trieschnigg, Dolf/Theune, Mariët: Folktale Classification Using Learning to Rank. In: *Advances in Information Retrieval. ECIR 2013. Lecture Notes in Computer Science*. ed. Pavel Serdyukov/Pavel Braslavski/Sergei O. Kuznetsov/Japp Kamps/Stefan Rüger/Eugene Agichtein/Ilya Segalovich/Emine Yilmaz. Heidelberg 2013, 195–206. https://doi.org/10.1007/978-3-642-36973-5_17.
- Pajupuu, Hille/Altrov, Rene/Pajupuu, Jaan: Identifying polarity in different text types. In: *Folklore. Electronic Journal of Folklore* 64 (2016) 125–142. <http://dx.doi.org/10.7592/FEJF2016.64.polarity>. <https://www.folklore.ee/folklore/vol64/polarity.pdf> (January 11, 2023).

- Pompeu, Duarte Pinto/Martins, Bruno/Matos, David: Interpretable Deep Learning Methods for Classifying Folktales According to the Aarne-Thompson-Uther Scheme. Master's Thesis, Instituto Superior Técnico, Universidade de Lisboa 2019.
- Propp, Vladimir: *Morphology of the Folktale*. New York 1958.
- Reiter, Nils/Frank, Anette/Hellwig, Oliver: An NLP-based cross-document approach to narrative structure discovery. In: *Literary and Linguistic Computing* 29,4 (2014) 583–605. <https://doi.org/10.1093/lilc/fqu055>.
- Seigneuret, Jean-Charles (ed.): *Dictionary of literary themes and motifs*. New York 1988.
- Silva, Sara G. da/Tehrani, Jamshid J.: Comparative phylogenetic analyses uncover the ancient roots of Indo-European folktales. In: *Royal Society Open Science* 3,1 (2016) 1–11. <https://doi.org/10.1098/rsos.150645>.
- Tangherlini, Timothy R.: Big folklore: A special issue on computational folkloristics. In: *The Journal of American Folklore* 129,511 (2016) 5–13. <https://doi.org/10.5406/jamerfolk.129.511.0005>.
- Tangherlini, Timothy R./Leonard, Peter: Trawling in the Sea of the Great Unread: Sub-corpus topic modeling and Humanities research. In: *Poetics* 41,6 (2013) 725–749. <https://doi.org/10.1016/j.poetic.2013.08.002>.
- Tehrani, Jamshid J.: The phylogeny of Little Red Riding Hood. In: *PLoS ONE* 8,11 (2013) e78871. <https://doi.org/10.1371/journal.pone.0078871>.
- Tehrani, Jamshid J./Nguyen, Quan/Rees, Teemu: Oral fairy tale or literary fake? Investigating the origins of Little Red Riding Hood using phylogenetic network analysis. In: *Digital Scholarship in the Humanities* 31,3 (2016) 611–636. <https://doi.org/10.1093/dsch/fqv016>.
- Thompson, Stith: *Motif-index of folk-literature: A classification of narrative elements in folktales, ballads, myths, fables, mediaeval romances, example, fabliaux, jest-books and local legends*. 2nd ed. Copenhagen 1951.
- Thompson, Stith: *The Folktale*. Berkeley 1977.
- Thuillard, Marc/d'Huy, Julien/Berezkin, Yuri Y./Le Quellec, Jean-Loïc: A Large-Scale Study of World Myths. In: *Trames Journal of the Humanities and Social Sciences* 22,4 (2018) 407–424. <https://doi.org/10.3176/tr.2018.4.05>.
- Uther, Hans-Jörg: *The Types of International Folktales: A Classification and Bibliography, Based on the System of Antti Aarne and Stith Thompson*. Helsinki 2004.
- Vapnik, Vladimir N.: An overview of statistical learning theory. In: *IEEE Transactions on Neural Networks* 10,5 (1999) 988–999. <https://doi.org/10.1109/72.788640>.
- White, John D.: *The Analysis of Music*. Prentice-Hall 1976.

Appendix: ATU tale type distribution of the AFT collection

The data can be accessed at doi.org/10.5281/zenodo.6575263

1. ANIMAL TALES (1–299)		46
Wild Animals 1–99		14
The Clever Fox (Other Animal) 1–69	1, 2, 15, 20C, 47A, 47B, 50, 57, 63, 66A, 68A	11
Other Wild Animals 70–99	75, 91, 92	3
Wild Animals and Domestic Animals 100–149	101, 103, 105, 112, 113A, 122E, 122F, 124, 130	9
Wild Animals and Humans 150–199	150, 154, 155, 156, 160, 173, 175, 178A	8
Domestic Animals 200–219	207C, 214A	2
Other Animals and Objects 220–299		
Birds 220–249	225, 231, 237, 243A, 244, 247	6
Fish 250–253		
Other animals and objects 275–299	275, 278, 278A, 280A, 285A, 295, 298	7
2. TALES OF MAGIC (300–749)		47
Supernatural Adversaries 300–399	303, 306, 310, 311, 312, 313, 325, 327, 328, 332, 333, 335, 361, 365, 366	15
Supernatural or Enchanted Wife (Husband) or Other Relative 400–459		6
Wife 400–424	402, 410	2
Husband 425–449	425C, 440, 441	3
Brother or Sister 450–459	451	1
Supernatural Tasks 460–499	480	1
Supernatural Helpers 500–559	500, 502, 503, 505, 510A, 510B, 545B, 555	8
Magic Objects 560–649	562, 563, 565, 570, 571B, 592, 613	7
Supernatural Power or Knowledge 650–699	650A, 670, 675	3
Other Tales of the Supernatural 700–749	700, 704, 706, 709, 720, 726, 737	7
3. RELIGIOUS TALES (750–849)		10
God Rewards and Punishes 750–779	750A, 756, 763, 777, 779, 779J*	6
The Truth Comes to Light 780–799	780, 782	2
Heaven 800–809	800	1
The Devil 810–826		
Other Religious Tales 827–849	845	1

4. REALISTIC TALES (NOVELLE) (850–999)	16
The Man Marries the Princess 850–869	850 1
The Woman Marries the Prince 870–879	875 1
Proofs of Fidelity and Innocence 880–899	882, 888 2
The Obstinate Wife Learns to Obey 900–909	900 1
Good Precepts 910–919	910B 1
Clever Acts and Words 920–929	920E, 926 2
Tales of Fate 930–949	
Robbers and Murderers 950–969	954, 955, 958E* 3
Other Realistic Tales 970–999	980, 980D, 981, 982, 990 5
5. TALES OF THE STUPID OGRE (GIANT, DEVIL) (1000–1199)	8
Labor Contract 1000–1029	
Partnership between Man and Ogre 1030–1059	1030 1
Contest between Man and Ogre 1060–1114	
Man Kills (Injures) Ogre 1115–1144	1137 1
Ogre Frightened by Man 1145–1154	
Man Outwits the Devil 1155–1169	1157, 1161 2
Souls Saved from the Devil 1170–1199	1174, 1175, 1176, 1191 4
6. ANECDOTES AND JOKES (1200–1999)	45
Stories about a Fool 1200–1349	1215, 1287, 1288A, 1317, 1319, 1335A, 1342, 1343 8
Stories about Married Couples 1350–1439	1351, 1353, 1362, 1365, 1377, 1381, 1381D, 1383, 1408, 1415, 1422, 1423, 1430 13
The Foolish Wife and her Husband 1380–1404	
The Foolish Husband and his Wife 1405–1429	
The Foolish Couple 1430–1439	1451 1
Stories about a Woman 1440–1524	
Looking for a Wife 1450–1474	
Jokes about Old Maids 1475–1499	
Other Stories about Women 1500–1524	
Stories about a Man 1525–1724	
The Clever Man 1525–1639	1540, 1548, 1558, 1562A, 1586, 1592, 1592B, 1620, 1626 9
Lucky Accidents 1640–1674	1641, 1641C, 1645, 1645B, 1655 5
The Stupid Man 1675–1724	1675, 1676, 1678, 1696 4
Jokes about Clergymen and Religious Figures 1725–1849	
The Clergyman Is Tricked 1725–1774	1730, 1741 2
Clergyman and Sexton 1775–1799	1791 1
Other Jokes about Religious Figures 1800–1849	
Anecdotes about Other Groups of People 1850–1874	
Tall Tales 1875–1999	1889B, 1965 2

7. FORMULA TALES (2000–2399)		10
Cumulative Tales 2000–2100	2015, 2022, 2025, 2030, 2031C, 2032, 2034F, 2035, 2043	9
Catch Tales 2200–2299	2250	1
Other Formula Tales 2300–2399		